Stylometric Analysis of Writing Style in Qwen Language Models

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Abstract

Large Language Models (LLMs) generate fluent and coherent text, but whether they exhibit consistent and distinguishable stylistic identities remains an open question. This study investigates whether three variants of the Qwen model family, namely Qwen2.5-7B, Qwen2.5-72B, and Qwen3-8B, produce texts with measurably distinct stylometric fingerprints. We collected 36 texts across four genres and extracted nine stylometric features. A Random Forest classifier achieved 63.9% accuracy under Leave-One-Out Cross-Validation (LOO-CV), nearly double the 33.3% chance baseline. Our results suggest that model size and generation affect writing style in quantifiable ways.

Keywords: Stylometry · Large language models · Authorship attribution · Natural language processing · Text analysis

1 Introduction

The emergence of Large Language Models (LLMs) has raised fundamental questions about the nature of machine-generated text. While much research has focused on the fluency, accuracy and reliability of LLM outputs, relatively limited attention has been placed on analyzing their aesthetic or stylistic properties. Do different models exhibit consistent "voices" and can these voices be quantified and distinguished?

Stylometry, the statistical analysis of writing style, has long been used in authorship attribution tasks. Recent work has extended these methods to AI-generated text [1], demonstrating that stylometric features can distinguish human from machine writing. Kumarage et al. (2023) demonstrated that stylistic signatures can identify AI-generated content even in short texts like tweets [2]. Zellers et al. (2019) showed that neural text generation can produce convincing fake news that closely mimics human writing, underscoring the importance of identifying model-specific stylistic fingerprints as a tool for detecting AI-generated content [3]. However, the question of whether different LLMs exhibit distinct styles from one another remains underexplored.

This study focuses on the Qwen model family developed by Alibaba, specifically three variants: Qwen2.5-7B-Instruct, Qwen2.5-72B-Instruct, and Qwen3-8B. These models were selected for three reasons. First, they are available on the Hugging Face free-tier API, ensuring reproducibility. Second, Qwen ranks among the top open-source LLMs globally, making it a valid subject for stylometric analysis. Third, the three variants differ in both size (7B to 72B parameters) and generation (Qwen2.5 vs Qwen3), enabling a controlled investigation of how these factors affect writing style.

2 Research Question and Methodology

2.1 Research Question

Our central research question is: *Do different variants of the same LLM family exhibit measurably distinct stylistic identities, and can these be quantified through stylometric analysis?*

We hypothesize that model size and generation will produce measurable differences in stylometric features, even within the same model family.

2.2 Dataset

We generated a corpus of 36 texts using three Qwen model variants accessed via the HuggingFace Inference API:

- **Qwen2.5-7B-Instruct** (qwen): small model, 7 billion parameters, second generation
- **Qwen2.5-72B-Instruct** (qwen72): large model, 72 billion parameters, second generation
- **Qwen3-8B** (qwen3): small model, 8 billion parameters, third generation

Each model was prompted with four genre-specific prompts, repeated three times each, yielding $3 \times 4 \times 3 = 36$ texts in total. Temperature was set to 0.7 to balance output diversity and consistency across repetitions. The four genres were:

- **Narration:** a short story about an orphaned child discovering a magic mirror.
- **Argumentation:** pros and cons of artificial intelligence in art and creativity
- **Dialogue:** a conversation between two old friends meeting after ten years
- **Description:** a sunset over the Mediterranean ocean

Keeping prompts constant across models ensures that any observed stylistic differences are attributable to the model rather than the topic.

2.3 Stylometric Features

We extracted nine stylometric features from each text, grouped into three categories following the framework proposed by Opara (2024):

Lexical features: Type-Token Ratio (lexical diversity), noun ratio, adjective ratio, verb ratio, measuring vocabulary richness, and part-of-speech distribution.

Syntactic features: Average sentence length, punctuation frequency, syntactic depth (ratio of subordinating conjunctions and relative pronouns), measuring sentence complexity and structural patterns.

Discourse features: Sentiment polarity (measured via TextBlob), Flesch Reading Ease score (measured via textstat), capturing affective tone and text accessibility.

2.4 Classification

To evaluate whether stylometric features can identify the source model, we trained a Random Forest classifier with 100 estimators. Given the small dataset size (36 texts), we used Leave-One-Out Cross-Validation (LOO-CV), training on 35 texts and testing on 1 in each iteration, to obtain unbiased accuracy estimates.

2.5 Dimensionality Reduction

To visualize stylistic distances between models and genres, we applied two complementary dimensionality reduction techniques. Principal Component Analysis (PCA) was used to identify the primary axes of stylometric variation. t-Distributed Stochastic Neighbor Embedding (t-SNE) was used to reveal local cluster structure, preserving neighborhood relationships in the high-dimensional feature space.

3 Experimental Results

3.1 Stylometric Feature Analysis

Table 1 presents the average stylometric features per model across all genres and repetitions.

Key observations from the feature analysis:

- **Qwen2.5-72B** produces the longest sentences (18.33 words/sentence) and uses the least punctuation (0.188), suggesting a more formal academic writing style.
- **Qwen3-8B** uses the most punctuation (0.285) and shortest sentences (15.54 words), particularly in dialogue, indicating a more expressive and dramatic style.
- **Qwen2.5-7B** exhibits the most balanced profile overall, positioned between the other two models on most features, with the exception of readability, where it scores slightly lower than both.

Table 1: Average stylometric features per model (across all genres and 3 repetitions)

Feature	Qwen2.5-7B	Qwen2.5-72B	Qwen3-8B
Lexical Diversity	0.704	0.664	0.710
Avg Sentence Length	16.06	18.33	15.54
Adjective Ratio	0.105	0.092	0.092
Verb Ratio	0.126	0.134	0.146
Noun Ratio	0.254	0.263	0.311
Punct Frequency	0.222	0.188	0.285
Syntactic Depth	0.086	0.093	0.075
Sentiment Polarity	0.178	0.193	0.118
Readability (Flesch)	57.63	58.51	58.47

3.2 Dimensionality Reduction

Figure 1 shows the PCA projection of nine stylometric features on two principal components. PC1 explains 97.2% of the total variance, with PC2 accounting for an additional 2.8%, together capturing the full stylometric variation in the dataset.

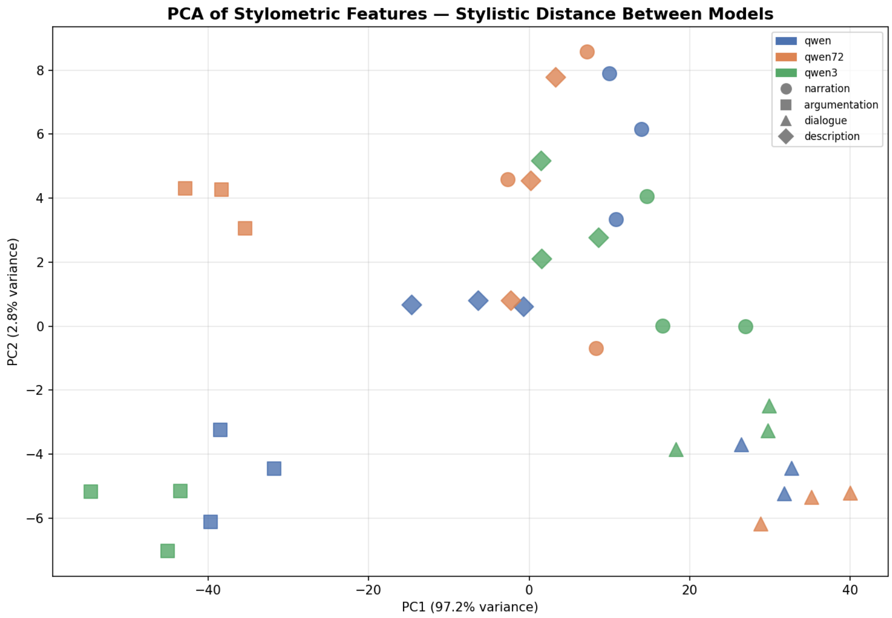

Figure 1: PCA of stylometric features. PC1 explains 97.2% of variance. Points represent individual texts; colors indicate model identity; shape indicate writing genre.

The PCA plot reveals that genre has a stronger separating effect than model identity. Argumentation texts (squares) form the most distinct cluster on the far left of the plot (PC1 negative), while dialogue texts (triangles) cluster on the far right (PC1 positive). Description texts (diamonds) occupy the central region, and narration texts (circles) are spread across the middle and right. Within each genre cluster, the three models show partial separation, confirming that model identity contributes a secondary but detectable stylometric signal.

Figure 2 presents the t-SNE visualization, which reveals clearer genre-based clustering than PCA. Argumentation texts (squares) form the tightest and most isolated cluster in the lower left region, while dialogue texts (triangles) are positioned in the upper right, furthest from all other genres. Description texts (diamonds) and narration texts (circles) occupy overlapping regions in the center, suggesting these two genres share more stylometric similarity.

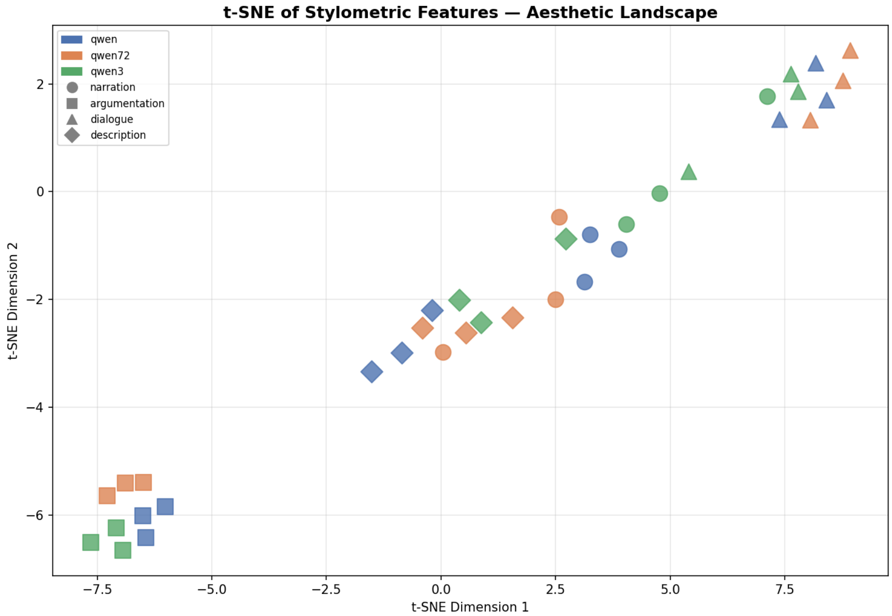

Figure 2: t-SNE aesthetic landscape of stylometric embeddings. Four distinct genre clusters emerge, with dialogue and argumentation showing the clearest separation.

Notably, within the dialogue cluster, the three models show clearer separation than in other genres, suggesting that model identity is most detectable in conversational writing.

3.3 Stylometric Feature Overview

Figure 3 presents all nine stylometric features in a single comparative visualization. Error bars indicate standard deviation across the three repetitions per genre, providing insight into the consistency of each model's stylometric profile.

Several patterns emerge from the overview. Across all models, dialogue consistently produces the highest punctuation frequency and lowest average sentence length, confirming that genre conventions strongly constrain stylometric output. Description texts achieve the highest lexical diversity, reflecting the richer vocabulary typically employed in scenic prose.

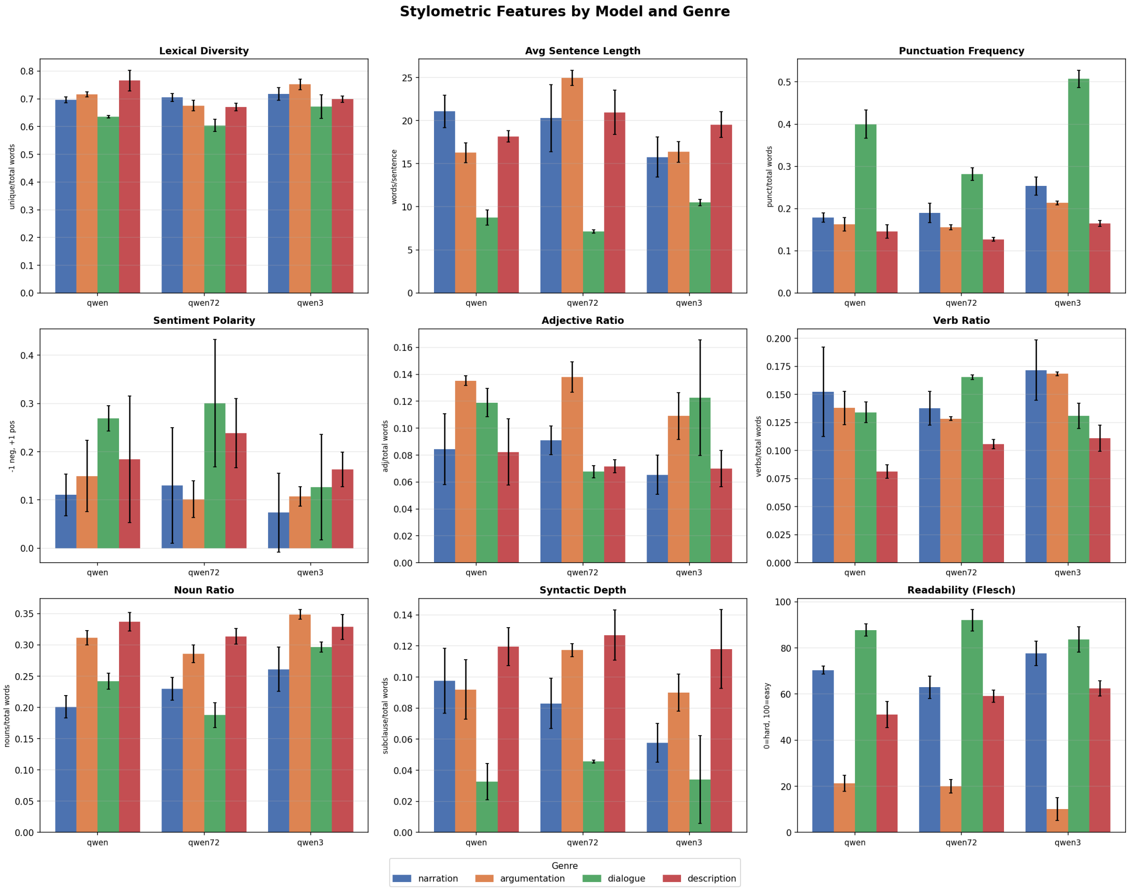

Figure 3: All nine stylometric features by model and genre. Error bars show standard deviation across 3 repetitions.

Regarding model-level differences, Qwen3-8B shows notably higher noun ratio (0.311) compared to the other two models (0.254 and 0.263), suggesting a more nominal writing style. Sentiment polarity varies most between models in the narration and description genres, where Qwen2.5-7B and Qwen2.5-72B tend toward more positive tone than Qwen3-8B. Readability scores vary significantly by genre rather than by model. Dialogue texts score highest (around 85–90) while argumentation texts score lowest (around 20), reflecting the inherent complexity of each genre rather than model-specific differences.

3.4 Classification Results

The Random Forest classifier achieved 63.9% accuracy under LOO-CV, nearly double the 33.3% chance baseline. Figure 4 shows the confusion matrix.

Qwen2.5-72B was the most reliably identified model (75% accuracy, 9 out of 12 correct), while Qwen2.5-7B was most frequently misclassified (50% accuracy, 6 out of 12 correct), suggesting its style is intermediate between the other two. Qwen3-8B achieved 67% accuracy (8 out of 12 correct).

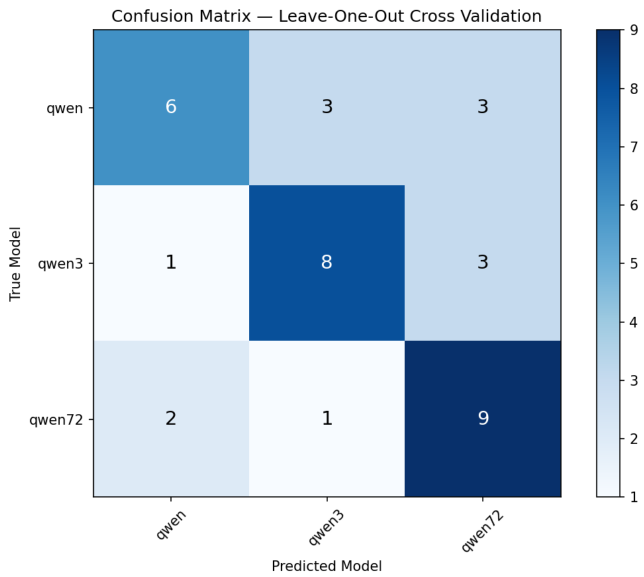

Figure 4: Confusion matrix from LOO-CV. Diagonal values represent correct predictions.

Notably, Qwen2.5-7B was equally confused with both other models (3 misclassifications each), while Qwen2.5-72B was rarely confused with Qwen3-8B (only 1 misclassification), indicating that model generation is a stronger stylistic differentiator than model size alone.

Figure 5 shows feature importance scores from the Random Forest classifier.

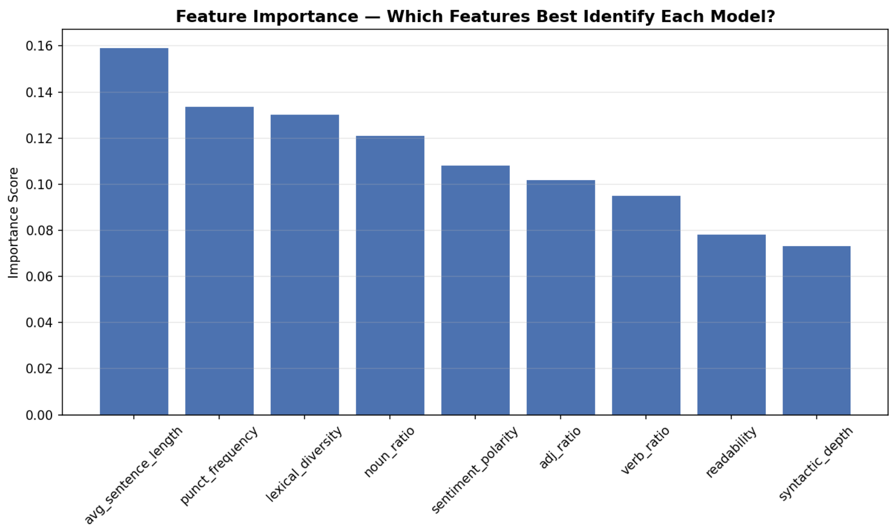

Figure 5: Feature importance scores from Random Forest classifier.

Average sentence length was the most discriminative feature (0.159), followed by punctuation frequency (0.133) and lexical diversity (0.130). Readability and syntactic depth contributed the least (0.078 and 0.073 respectively), suggesting these features carry limited discriminative power for distinguishing Qwen model variants.

3.5 Style Transfer Experiment

As an optional extension, we investigated whether stylometric features can be deliberately transferred between models. We instructed Qwen2.5-7B to rewrite its own texts in the formal academic style characteristic of Qwen2.5-72B, repeating each rewrite three times and averaging the results for reliability. A feature was considered successfully transferred if its value moved closer to the Qwen2.5-72B average after rewriting.

Table 2 summarizes the style transfer success rate per genre.

Table 2: Style transfer success rate: features shifted toward Qwen2.5-72B profile (averaged over 3 runs)

Genre	Features Shifted	Success Rate
Narration	3/9	33.3%
Argumentation	6/9	66.7%
Dialogue	3/9	33.3%
Description	4/9	44.4%
Overall	**16/36**	**44.4%**

Overall, 44.4% of stylometric features shifted toward the Qwen2.5-72B profile across all genres. Results varied by genre: argumentation achieved the highest transfer rate (6/9 features, 66.7%), while narration and dialogue showed the lowest (3/9 features, 33.3% each). Syntactic features such as verb ratio and noun ratio were most successfully transferred across genres, while surface features including readability and punctuation frequency proved most resistant to explicit style instructions, suggesting that deep syntactic style is more malleable than surface-level markers.

4 Concluding Remarks

This study demonstrates that variants of the Qwen model family exhibit measurably distinct stylometric fingerprints. A Random Forest classifier trained on nine stylometric features achieves 63.9% accuracy in identifying the source model under LOO-CV, nearly double the 33.3% chance baseline. These results confirm that model size and generation affect writing style in quantifiable ways.

Our analysis reveals four key findings. First, genre has a stronger effect on stylometric features than model identity, as shown by the clear genre-based clustering in both PCA and t-SNE visualizations. Second, Qwen2.5-72B exhibits the most distinctive style (longest sentences, least punctuation), suggesting that larger models adopt a more formal register. Third, Qwen3-8B displays the most expressive style (most punctuation, shortest sentences), particularly in dialogue, indicating that model generation influences stylistic choices beyond mere scale. Fourth, model identity is the most detectable in dialogue writing, where the three models show the clearest within-genre separation.

The style transfer experiment provides additional evidence for the robustness of these stylometric signatures. When explicitly instructed to adopt Qwen2.5-72B's formal style, Qwen2.5-7B shifted only 44.4% of features toward the target profile, suggesting that some stylometric properties are deeply embedded in the model's generation process and resistant to surface-level prompting.

Future work should address the main limitations of this study, including the small dataset size (36 texts) and the exclusive use of the Qwen model family, by incorporating more diverse model families and larger corpora.

AI Usage Disclaimer

Parts of this project were developed with the assistance of Claude (Anthropic). The AI was used to support code development and prompt design. All content has been reviewed, edited, and validated by the author. Full responsibility for the content rests with the author.

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